

Fuzzy Membership-Based Ensemble Learning for Robust Multilingual Speech Emotion Recognition

1st Dr. Poonam Rani
CSE Department
NSUT

Dwarka, New Delhi, India
poonam.rani@nsut.ac.in

2nd Anshika Verma
CSE Department
NSUT

Dwarka, New Delhi, India
anshika.verma.ug23@nsut.ac.in

3rd Ekta
CSE Department
NSUT

Dwarka, New Delhi, India
ekta.ug23@nsut.ac.in

4th Jiyaa Gupta
CSE Department
NSUT

Dwarka, New Delhi, India
jiyaa.gupta.ug23@nsut.ac.in

5th Chahat

CSE Department
NSUT

Dwarka, New Delhi, India
chahat.ug23@nsut.ac.in

Abstract—Speech Emotion Recognition (SER) is an important area in affective computing and human-computer interaction. This paper proposes a fuzzy cluster-guided ensemble framework for multilingual SER using RAVDESS and SUBESCO emotional speech datasets. The framework integrates acoustic feature extraction, Principal Component Analysis (PCA), Fuzzy C-Means (FCM) clustering, fuzzy membership-based feature augmentation, and ensemble machine learning techniques for accurate emotion classification. Acoustic features such as MFCC, Mel Spectrogram, Chroma Features, Zero Crossing Rate, Spectral Bandwidth, and Pitch were extracted to represent emotional characteristics. PCA reduced feature redundancy, while FCM modeled overlapping emotional distributions through soft membership assignment. Multiple classifiers, including Random Forest, SVM-RBF, XGBoost, and CatBoost, were evaluated using ensemble voting strategies. Experimental results achieved 84.54% accuracy and 84.51% weighted F1-score, demonstrating the framework’s effectiveness and robustness for multilingual SER applications.

Index Terms—Speech Emotion Recognition (SER), Fuzzy C-Means Clustering, Ensemble Learning, PCA, MFCC, Multilingual Speech Processing, and Emotion Classification.

I. INTRODUCTION

Human emotions are an integral part of natural communication that affect the expression of thoughts, intentions, attitudes, and psychological states. Of the modes of expression, speech is one of the most direct and natural modes as it not only carries the spoken message but also includes a wealth of vocal cues including pitch, tone, energy, rhythm, intensity, pauses and speaking rate. Many of these acoustic features will indicate the emotional state of the speaker even though the words used are in themselves neutral. Hence, the field of Sentiment Speech Recognition (SER) has become of great significance in Affective Computing and Human-Robot Interaction (HRI) and Human-Computer Interaction (HCI).

The goal of SER is to automatically recognise the emotional states expressed by speech signals and classify them as happiness, sadness, anger, fear, neutral, disgust or surprise. Due to its versatile applications across the healthcare sector, e-learning, call centers, virtual assistants, gaming, driver monitoring, and mental health support systems, it has become more significant. For instance, an SER-based system could tell when a user is stressed or frustrated and adjust how the machine operates; the machines could adapt and become more user-friendly based on a user’s voice stress.

Although it is a critical task, emotional speech is challenging due to the diversity of emotions, languages, accents, genders, age groups and cultures. Further, recognition accuracy can be influenced by background noise, the small emotional sample size, class imbalance and by the fact that emotional expressions are sometimes similar. Most traditional SER approaches relied on hand-crafted acoustic features and traditional machine learning classifiers.

In recent years, however, deep learning has made it possible to automatically learn discriminative temporal and spectral patterns from speech signals. The CNN, RNN, LSTM, GRU, and Transformer-based models have demonstrated great promise in enhancing emotion classification accuracy.

In this paper an experimental Speech Emotion Recognition system is implemented to analyze emotional information contained in speech signals. This study is centered on extracting meaningful acoustic features and using learning based classification methods to enhance the recognition of the various emotional states. The goal is to develop a reliable and efficient model of SER which will help develop more emotion-aware, intelligent, human-computer interaction systems.

II. LITERATURE REVIEW

Several studies have been conducted for the crucial role of the Speech Emotion Recognition (SER) and the Affective Computing and Human Computer Interaction (HCI) with different acoustic characteristics, classification methods and deep learning architectures which have been investigated to enhance the emotional recognition from speech signals. The SER systems used prior to using handcrafted acoustic features and traditional classifiers were mostly used. The set of features like MFCC, DWT, pitch, energy, Zero Crossing Rate are extracted and features are selected and classified by [1]. The use of proper speech characteristics and the performance of speech recognition were directly related to the accuracy, achieved with 85% with the use of Decision Tree classifiers when using the proper speech characteristics. Also, [2] computed the MFCC for every speech frame and voted for the final emotion label according to the maximum number of votes for the particular emotion label, which achieved the best accuracy (70%). This has shown that low dimensional acoustic features could be a good solution in the emotion of speech classification.

The machine learning has also evolved and optimized models of classification have been developed for SER as well. [3] proposed an Improved Weighted Average Support Vector method with Reptile search optimization (IWASV-RS) to select features, by combining the Reptile search optimization method with the Improved Weighted Average Support Vector method. From their optimized SVM based models it has been proved that optimized SVM based models can give good performance in the task of Speech Emotion Classification with 98.56 % level of accuracy of the optimized SVM based models. To further enhance the accuracy of MFCC feature, [4] fused the existing MFCC features with the spectrograph features obtained by a CNN model previously trained. They used their MFCC-LSTM model and found that they could reach 73.5% accuracy, while the combination of classical acoustic features (mfcc-spectrogram) and SVM could reach 71.3% accuracy, indicating that classical acoustic features are still useful even when combined with the deep learning based features.

More and more focus has been drawn to deep learning based architectures in the recent years due to their potential of automatic learning of complex temporal and spectral patterns of speech. In the current study by [5] the authors created an lightweight model of 1D-CNN which embeds the combined features of MFCC, Mel-spectrogram and Chroma. They achieved 94.0% accuracy on MELD and 91.9% accuracy on RAVDESS and in their model, they realized that the feature fusion can be done without introducing many features, and thus improve the accuracy of the audio-only SER. [6] present an innovative parallel Deep Convolutional Network (DCN) with features of Multitaper Mel Frequency Spectrogram (MFMS), Gammatonegram (GTG) and Constant-Q Transform (CQT). Their method improved the distinction of the emotional speech features in multiple spectral representations with 97.14% and 97.41% accuracy on the two databases

EMODB and RAVDESS respectively, which indicates their method can be used to effectively distinguish the emotional speech features in multiple spectral representations.

In addition, Sequential learning and self-supervised learning method have also been playing a significant role in SER. For only acoustics [7] used Wav2Vec embeddings as the embedder, and then stacked deeper GRU layers for that task, resulting in 79.82% accuracy on RAVDESS. Based on their research, they draw a conclusion that their pre-trained speech representations and recurrent models have learned temporal patterns of emotion. [8] emphasizes the need to fix the problem of the imbalance nature of the dataset and use a deep feature fusion method to achieve better generalization with the application of a Balanced Focal Entropy loss function. Finally, the model performed 95.23% accuracy on the RAVDESS dataset and 97.80% accuracy on the hybrid dataset, showing how crucial the diversity of datasets and loss optimization is for SER.

One of the main topics in the past few years in SER is temporal dependency modelling. [9] suggested a Multi-scale Temporal Causal Network (MTCN) to learn multi-scale speech features and long-range emotional dependencies. Their model achieved 96.64% accuracy on EMODB, 93.53% on EMOVO, 93.13% on RAVDESS, 91.25% on SAVEE, 95.17% on CASIA, and 71.33% on IEMOCAP. The summary of all the studies conducted shows that the acoustic features quality, feature fusion, optimization of the classifier, modelling the dynamics over time and diversity of the data set are important factors which influence the SER performance. There remain issues, including class imbalance, similarity of different emotions, limited number of labelled data, speaker variation and noise (background noise, movement) that will be our driving forces for further experiments with SBER.

III. MATERIAL AND METHOD

A. Dataset Description

In this study we looked at how to recognize emotions in speech. We used two collections of emotional speech to do this. These collections are called the RAVDESS dataset and the SUBESCO dataset. They have lots of clips of people expressing different emotions like anger, happiness, sadness, fear, disgust and surprise.

The RAVDESS dataset has good quality recordings of people speaking with different emotions. The SUBESCO dataset has a lot of speech clips in Bengali, which's a language spoken in some parts of the world.

We used a total of 8440 speech samples for this study. These samples are divided into eight groups based on the emotion they express. We figured out which emotion each sample expressed by looking at the name of the file.

The emotions we looked at are:

- Angry
- Calm
- Disgust
- Fear
- Happy

- Neutral
- Sad
- Surprise

We think the dataset is evenly balanced meaning we have a similar number of samples, for each emotion. This balance is helpful because it makes it easier to get results when we try to classify the emotions.

B. Audio Preprocessing

The voice data was cleaned up before we looked for any features to make sure we got a good sense of the emotions in the audio clips. To clean up the voice data we did these things:

- Loaded the audio signals
- Removed noise from the signal
- Made sure all the signals were at the level
- Broke the signals into parts and looked at them one by one
- Looked for the features

We used Python and some special helpers, like Librosa and NumPy to work with the audio signals.

C. Feature Extraction

One of the steps in the Speech Emotion Recognition process is finding the right features. For this project we looked at the things that show emotions in peoples voices.

We found features, like:

- Mel Spectrogram
- Mel Frequency Cepstral Coefficients
- Zero Crossing Rate
- Chroma Features
- Spectral Bandwidth
- Pitch which's the basic frequency of a persons voice.

We saved these features in two formats, CSV and NumPy so we can use them later in the Speech Emotion Recognition process.

1) *Mel Spectrogram*: Mel Spectrogram shows the distribution of speech signals on the Mel scale, which reflects the human auditory system closely. The conversion of frequency to the Mel scale makes it possible to detect emotional patterns from speech.

The Mel frequency conversion formula is:

$$m = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

Where:

- m = Mel frequency
- f = Original frequency in Hz

Mel Spectrogram features help identify variations in pitch, intensity, and speaking style associated with emotions.

2) *Mel Frequency Cepstral Coefficients (MFCC)*: MFCC is one of the most widely used features in speech processing and emotion recognition. It captures important spectral characteristics of speech based on human hearing perception.

The MFCC extraction process includes:

Pre-emphasis Framing Windowing Fast Fourier Transform (FFT) Mel filter bank processing Logarithmic scaling Discrete Cosine Transform (DCT)

The MFCC coefficient equation is:

$$MFCC(n) = \sum_{m=1}^M \log(S_m) \cos \left[\frac{\pi n(m - 0.5)}{M} \right]$$

Where:

- S_m = Mel filter bank energies
- M = Number of Mel filters
- n = MFCC coefficient index

In this work, 20 MFCC coefficients were extracted from each speech sample. MFCC features help represent vocal tract characteristics and emotional speech variations.

3) *Zero Crossing Rate (ZCR)*: The Zero Crossing Rate indicates how often the speech signal crosses the zero amplitude axis. It aids in recognizing the emotional intensity and signal noisiness.

The ZCR formula is:

$$ZCR = \frac{1}{2N} \sum_{n=1}^N |\text{sign}(x_n) - \text{sign}(x_{n-1})|$$

Where:

- x_n = Speech signal amplitude
- N = Total number of samples

High ZCR values are often associated with emotions such as anger or excitement.

4) *Chroma Features*: Chroma features represent the distribution of frequency content across 12 different pitch classes. Chroma features play a role in capturing the harmonic and tonal aspects of the speech signal.

Chroma-based features are a useful approach for identifying changes in emotional tone and pitch.

5) *Spectral Bandwidth*: The spread of frequencies around the spectral centroid is measured by spectral bandwidth. It shows how frequencies are arranged in the speech signal. The formula for spectral bandwidth is: It shows the distribution of frequencies within the speech signal.

The equation for spectral bandwidth is:

$$\text{Bandwidth} = \sqrt{\sum_{k=1}^N (f_k - c)^2 P(f_k)}$$

Where:

- f_k = Frequency component
- c = Spectral centroid
- $P(f_k)$ = Power spectrum

Spectral bandwidth helps distinguish between calm and highly expressive emotions.

6) *Pitch (Fundamental Frequency)*: Fundamental speech frequency is indicated by pitch, which is crucial for recognizing emotions. When a speech is emotional, it often leads to changes in pitch level.

The formula for calculating pitch frequency is:

$$F_0 = \frac{1}{T}$$

Where:

- F_0 = Fundamental frequency
- T = Time period of the waveform

Emotions like surprise and happiness are usually linked to a high pitch, whereas a low pitch can signify sadness or calmness.

IV. PROPOSED WORK

The proposed Speech Emotion Recognition (SER) approach involves the utilization of a hybrid approach based on Fuzzy C-Means (FCM)-inspired ensemble learning methodology. The process involves feature extraction, dimensionality reduction, fuzzy clustering, and an ensemble-based learning approach for identifying emotions in speech samples.

The main steps involved in the processing pipeline of the proposed approach include:

- 1) Input Audio
- 2) Preprocessing
- 3) Feature Extraction
- 4) Feature Normalization
- 5) Principal Component Analysis (PCA)
- 6) Fuzzy C-Means (FCM) Clustering
- 7) Feature Augmentation Based on Membership
- 8) Training of Ensemble Classifier
- 9) Prediction Through Voting

A. Audio Input and Preprocessing

The audio data from the speech is initially collected from the databases of emotional speech. The audio data is standardized and processed before the features can be extracted.

The initial processing phase entails:

- Noise filtering
- Normalization of the signal
- Framing
- Windowing

This process ensures that the quality of the speech data improves through this stage.

B. Feature Extraction Stage

After preprocessing we pull out acoustic and spectral features from speech signals. These features help us understand the emotions in speech.

The features we pull out are:

- Mel Spectrogram
- MFCC
- Zero Crossing Rate or ZCR for short
- Chroma Features
- Spectral Bandwidth

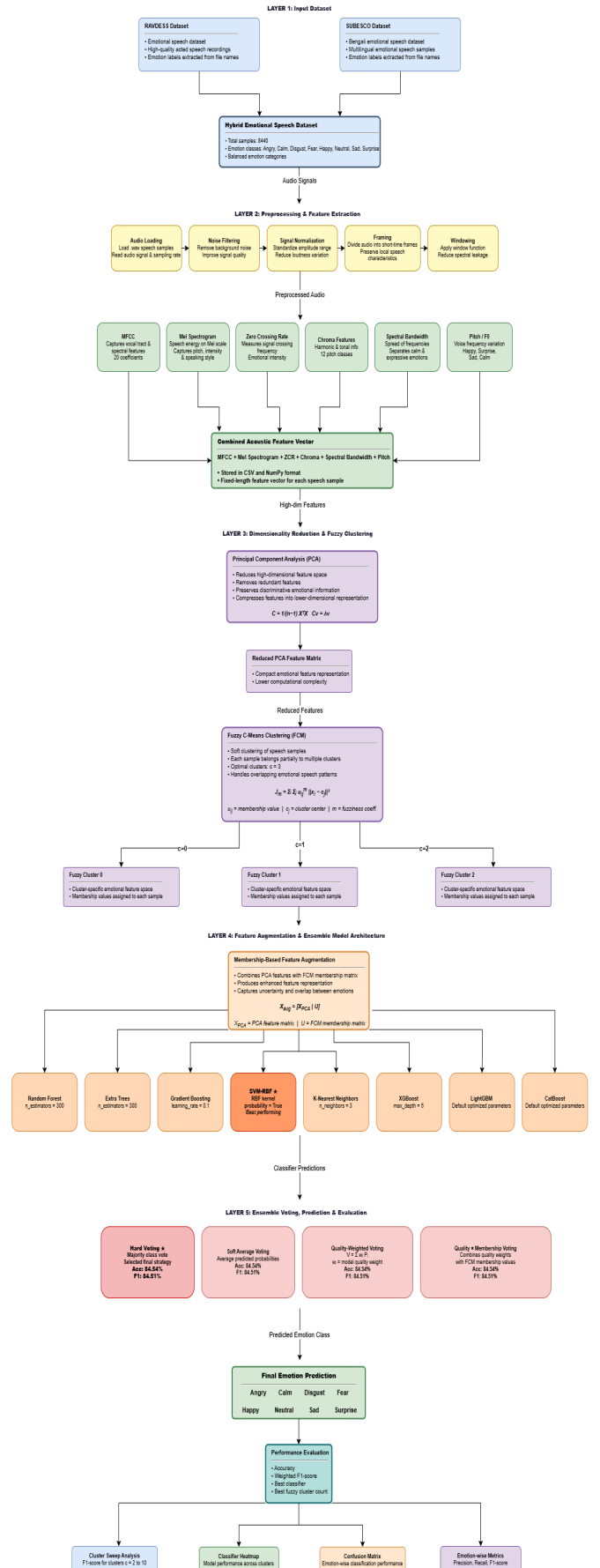


Fig. 1: Workflow Architecture

- Pitch Features

We then combine these features into a feature vector of a fixed length, for each speech sample. This helps us analyze the speech data.

C. Principal Component Analysis (PCA)

The extracted feature vectors are high-dimensional and may contain redundant information. Therefore, PCA is applied to reduce dimensionality while preserving important emotional information.

The covariance matrix used in PCA is:

$$C = \frac{1}{n-1} X^T X$$

The eigenvalue decomposition equation is:

$$Cv = \lambda v$$

Where:

- C = Covariance matrix
- v = Eigenvector
- λ = Eigenvalue

D. Fuzzy C-Means (FCM) Clustering

Fuzzy C-Means clustering is used to group similar speech samples. It does this in a way, than hard clustering. Fuzzy C-Means clustering gives each speech sample a score that shows how well it fits into groups.

The Fuzzy C-Means objective function is:

$$J_m = \sum_{i=1}^N \sum_{j=1}^C u_{ij}^m \|x_i - c_j\|^2$$

Where:

- u_{ij} = Membership value
- m = Fuzziness coefficient
- x_i = Input feature vector
- c_j = Cluster center

The membership update equation is:

$$u_{ij} = \frac{1}{\sum_{k=1}^C \left(\frac{\|x_i - c_j\|}{\|x_i - c_k\|} \right)^{\frac{2}{m-1}}}$$

E. Membership-Based Feature Augmentation

The fuzzy membership values, from FCM are mixed with PCA features to make feature vectors.

The new feature vector looks like this:

$$X_{aug} = [X_{PCA} \mid U]$$

Where:

- X_{PCA} = PCA feature matrix
- U = Membership matrix

F. ML Classification

Multiple machine learning models are trained independently on the augmented feature space.

The classifiers used are:

- Random Forest
- Extra Trees
- Gradient Boosting
- SVM-RBF
- KNN
- XGBoost
- LightGBM
- CatBoost

Cross-validation is used to evaluate model performance and select the best classifier for each cluster.

The Support Vector Machine (SVM-RBF) achieved the highest classification performance in the proposed system.

G. Quality-Weighted Voting Strategy

The outputs from multiple classifiers are combined using ensemble voting strategies.

The voting approaches include:

Hard Voting Soft Voting Quality-Weighted Voting
Membership-Weighted Voting

The quality-weighted voting score is calculated as:

$$V = \sum_{i=1}^C w_i P_i$$

Where:

- w_i = Model quality weight
- P_i = Predicted probability from classifier

The final system achieved:

Accuracy = 84.54% Weighted F1-score = 84.51%

V. RESULT

This study is aimed at evaluating the proposed fuzzy cluster guided ensemble framework for Speech Emotion Recognition (SER) with a hybrid dataset of 8440 emotional speech samples from RAVDESS and SUBESCO. The suggested framework has been based on Principal Component Analysis (PCA), Fuzzy C-means (FCM) Clustering, Membership weighted ensemble learning and multiple machine learning classifiers. Eight classifiers namely Random Forest (RF), Extra Trees (ET), Gradient Boosting (GB), Support Vector Machine with RBF kernel (SVM-RBF), K-Nearest Neighbors (KNN), XGBoost (XGB), LightGBM (LGBM), and CatBoost were used for the experimental analysis.

Table 1 shows the best set of parameters used for the classifiers evaluated. An optimization process for the number of fuzzy clusters was conducted varying the number of clusters between 2 and 10 and judging the downstream weighted F1 score. The best results are achieved for 3 fuzzy clusters ($c = 3$) and a weighted F1-score of 0.6895 in the cluster sweep stage.

The proposed framework achieved the highest classification accuracy of 84.54 % and the weighted F1 score of 84.51 % on

TABLE I: Optimal Parameters of Machine Learning Models

Model	Optimal Parameters
SVM-RBF	RBF kernel, probability=True
Random Forest	n_estimators = 300
Extra Trees	n_estimators = 300
Gradient Boosting	learning_rate = 0.1
KNN	n_neighbors = 3
XGBoost	max_depth = 5
LightGBM	default optimized parameters
CatBoost	default optimized parameters

the validation set with SVM-RBF Classifier. The performance comparison of the different voting strategies for getting the desired results in the validation is given in Table 2. The overall best strategy from the voting strategies analyzed was the hard voting.

TABLE II: Performance Comparison of Different Voting Strategies

Strategy	Accuracy	Weighted F1-score
Hard Voting	0.8454	0.8457
Soft Average Voting	0.8411	0.8414
Quality-weighted Voting	0.8454	0.8455
Quality $\times$ Membership Voting	0.8454	0.8454

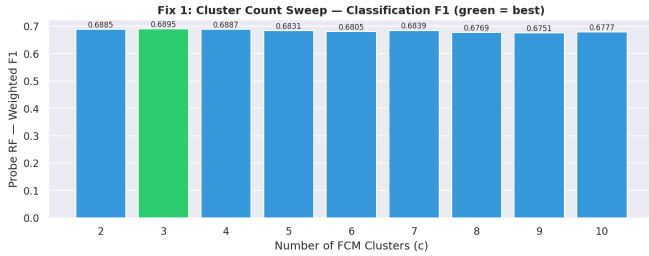


Fig. 2: F1-score analysis for different cluster configurations.

Table 3 provides the performance comparison of all the classifiers on the fuzzy clusters. SVM-RBF was the model with the best validation F1-score among all considered models, 0.8347, in all the clusters. Extra Trees and Random Forest also showed competitive results, and KNN and CatBoost showed comparatively low scores.

TABLE III: Classifier Performance Across Clusters

Classifier	Cluster 0	Cluster 1	Cluster 2
Random Forest	0.7729	0.7737	0.7706
Extra Trees	0.7982	0.7941	0.7969
Gradient Boosting	0.7006	0.6993	0.6994
SVM-RBF	0.8135	0.8135	0.8135
KNN	0.6768	0.6768	0.6768
XGBoost	0.7407	0.7438	0.7420
LightGBM	0.7517	0.7508	0.7508
CatBoost	0.6534	0.6537	0.6558

The final classification report of the proposed framework is given in Table 4. The model had good emotion classi-

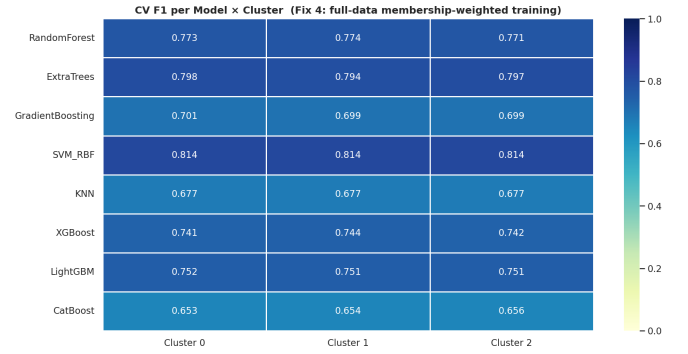


Fig. 3: Heatmap representation of model performance across clusters.

fication accuracy for most emotion classes. The class with the maximum number of training samples (angry) achieved the highest score whereas the comparatively low number of training samples in the class of calm emotion resulted in comparatively lower scores.

TABLE IV: Emotion-wise Performance Metrics

Emotion	Precision	Recall	F1-score
Angry	0.88	0.91	0.89
Calm	0.70	0.68	0.69
Disgust	0.81	0.81	0.81
Fear	0.89	0.81	0.85
Happy	0.87	0.86	0.86
Neutral	0.86	0.90	0.88
Sad	0.85	0.86	0.85
Surprise	0.80	0.79	0.80

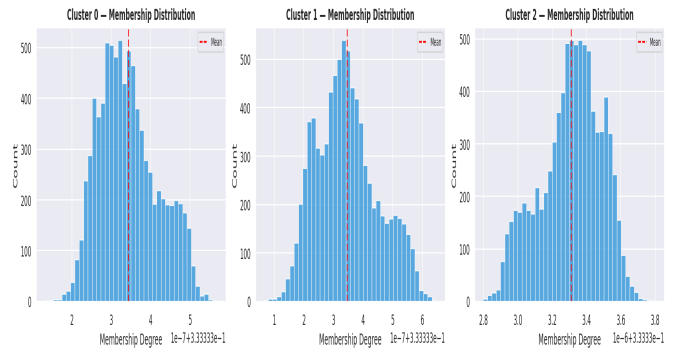
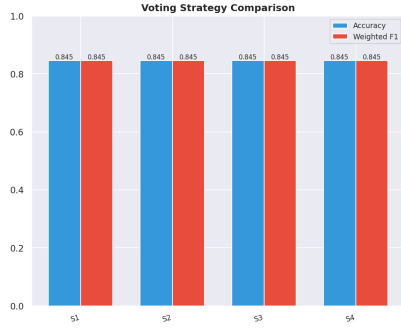


Fig. 4: Distribution of membership weights for the selected features.

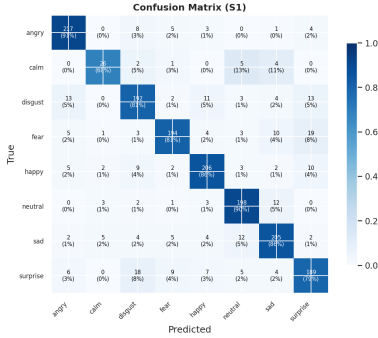
The results of the cluster membership analysis indicated that the fuzzy memberships were very close for each sample. Although the membership of the partitions had been fuzzy leading to overlaps in membership, the performance of classification was still satisfactory even in the framework of the proposed ensemble. The feature compression method using PCA extracted 17-dimensional features from 100-dimensional original features, removing redundant and redundant features,

such that the extracted features represent the most discriminative information related to each emotion.

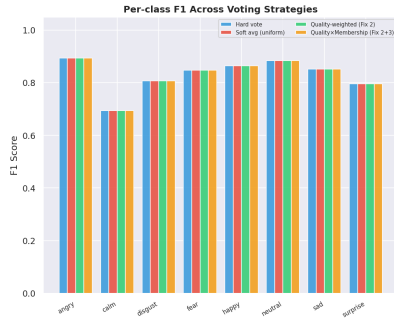
The cluster sweep analysis based on weighted F1-Score with various number of clusters is shown in Fig. 2. The results of the model-cluster are shown in a heatmap in Fig. 3, which shows the performance of each of the classifiers in each cluster. The membership weight distributions computed by the FCM clustering process are presented in Fig. 4 and the final classification result as well as the confusion matrix can be seen in Fig. 5.



(a) Voting Strategy Comparison



(b) Confusion Matrix



(c) Per-Class F1 Across Voting Strategies

Fig. 5: Final performance comparison of all evaluated classifiers.

VI. DISCUSSION

The proposed fuzzy cluster guided ensemble framework is better compared to the other methods, which show that the PCA based dimensionality reduction, Fuzzy C-Means

(FCM) clustering and ensemble machine learning techniques are integrated into the proposed framework well and it shows strong and reliable performance for multilingual Speech Emotion Recognition (SER). The framework overall validated at 84.54% and weighted F1 at 84.51% indicating that the framework has good capability on emotional discrimination in the eight emotion classes. The outcomes achieved demonstrate the validity of the proposed fuzzy clustering approach with ensemble learning technique for the highly overlapping and fuzzy emotional speech signals.

The results of experimental analysis show that in almost all fuzzy clusters SVM-RBF classifier outperformed all other classifiers that were evaluated. This illustrates the effectiveness of non-linear kernel-based learning in learning the complex emotional boundary in the low dimensional PCA feature space. Moreover, ensemble classifiers such as Random Forest (RF) and Extra Trees (ET) have also performed well and demonstrated the robustness and stability of the tree based ensemble classifiers in multilingual SER problems.

Main benefit in the suggested framework is the fuzzy membership guided learning technique. The proposed method does not follow the traditional hard clustering method, where a speech sample is completely placed into one emotional cluster, as it is possible to have a speech sample to be partially put into several emotional clusters simultaneously. The soft assignment mechanism significantly improves the robustness of emotion classification, which shares acoustic characteristics and has a similar feature in the emotional content, for instance, fear, sadness, and surprise. Furthermore, optimization of the cluster using downstream weighted F1 score can be used for better generalization of the framework and specialization of classifiers in each cluster.

Several advantages are suggested for the proposed framework, compared with other SER approaches. Most previous work has been focussed on voting based SER systems and [1] on optimization of feature extraction technique. Likewise, [6] suggested the deep learning based emotional representation learning architectures and [3] introduced weighted SVM optimization strategies for SER. Most of these methods, however, require time consuming computationally deep learning architectures and/or high grade of handcrafted optimization.

The proposed scheme, however, combines fuzzy clustering and membership-based ensemble voting, PCA-based feature compression and classifier optimization together in an efficient SER pipeline. This integration allows the framework to have good classification results with a lower complexity and higher interpretability than deep neural architectures.

[10], [5], [9], and [7] recently introduced deep learning methods, such as CNNs, temporal causal networks, transformer embeddings, and Wav2Vec architectures, that had high SER performance. These models typically have a high parameter count, require a large amount of tuning of parameters and (very large) training sets, and have satisfactory performance. By comparison, a proposed framework achieves competitive performance with lightweight machine learning models, and significantly reduces the computational requirements. This

makes the proposed approach more practical and scalable in the real SER application, where computation resources is limited.

A. Comparison with State-of-the-Art Work

Table 5 presents the comparison between the proposed framework and recent SER approaches reported in the literature. The proposed framework achieved competitive performance while utilizing comparatively lightweight machine learning models and fuzzy ensemble learning.

TABLE V: Comparison with Existing State-of-the-Art Methods

Reference	Method	Dataset	Accuracy
[2]	MFCC + LMT with voting mechanism	Emotional speech dataset (7 emotions)	70%
[1]	Feature extraction + Decision Tree / SVM / LDA	Multiple datasets	85% (DT), 70% (SVM), 65% (LDA)
[4]	MFCC-LSTM / Hybrid SVM (MFCC + Spectrogram)	Speech emotion datasets	73.5% / 71.3%
[7]	Wav2Vec + Deep GRUs	RAVDESS	79.82%
[10]	CNN-based SER Model	RAVDESS	80.00%
Proposed Framework	PCA + FCM + Ensemble Learning	RAVDESS + SUBESCO	84.54%

The proposed framework is also generalizable on multilingual emotional speech datasets. The feature redundancy problem can be reduced while preserving the most discriminative emotional information by using the combination of PCA and fuzzy clustering. Moreover, the ensemble voting strategy yields more stable prediction and reduces the variability of classifiers, thus leading to improved emotion recognition performance.

There are a few drawbacks in the current system, however. The fuzzy membership distributions resulting from experimentation were fairly equally distributed for some samples suggesting that the cluster structure in the emotional feature space is overlapping. This implies some emotions have similar acoustic characteristics and can make the cluster separability difficult. Future directions involve the adaptive fuzzy clustering strategies, attention-guided feature learning and deep fuzzy representation learning for improving the emotional discrimination and clustering compactness.

Overall, the experimental results prove that the proposed fuzzy cluster-guided ensemble approach is effective, interpretable and computationally efficient for MSEER. The framework is able to achieve the balance between the classification accuracy, robustness, computational efficiency and generalization ability, which is better than several existing traditional and deep learning based SER approaches.

VII. CONCLUSION AND FUTURE WORK

This study presented a fuzzy cluster-guided ensemble framework for multilingual Speech Emotion Recognition (SER) using hybrid emotional speech datasets. The proposed system integrates acoustic feature extraction, PCA-based dimensionality reduction, Fuzzy C-Means clustering, and ensemble machine learning classifiers to improve emotion classification performance. Fuzzy membership learning effectively captures the inherent uncertainty and overlap in emotional speech, enabling better discrimination between similar emotional states.

The use of PCA reduces feature redundancy while preserving essential emotional information, resulting in improved computational efficiency and stable predictions. Experimental evaluation achieved an accuracy of 84.54% and a weighted F1-score of 84.51%, with the SVM-RBF classifier demonstrating the best performance. Overall, the framework shows strong interpretability and generalization capability across multilingual datasets.

Future work will focus on enhancing adaptive fuzzy clustering mechanisms and incorporating advanced deep learning architectures such as Transformers and attention-based models to further improve recognition performance. Additionally, integrating multimodal emotional cues, including facial expressions and textual sentiment, may strengthen system robustness. Further research can also explore real-time deployment, noise-robust modeling, lightweight optimization strategies, and validation on large-scale cross-cultural datasets to improve practical applicability in domains such as healthcare systems, virtual assistants, and intelligent human-computer interaction systems.

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